

Variable Speed Drive (VSD) Motor MAPS Template (Electrical)

The Variable Speed Drive (VSD) Motor MAPS template is designed to control a VSD motor that is connected by five outputs one for forward direction and one for reverse direction and three for the speed selection.

All MAPS templates provide the following:

- an associated PLC function block
- a set of SCADA tags (Adroit agents) linked to the provided signals
- an associated SCADA graphic with inbuilt faceplates

This document describes the Advanced Variable Speed Drive (VSD) Motor MAPS template. For details of its required PLC memory usage; number of PLC I/O and SCADA size (scan points) - see the **Resources** table below.

Advanced VSD Motor Functional Philosophy

The VSD start forward and reverse direction outputs are enabled once all interlocks are ready and a start request is received.

Field Interlocks

- MCC (Overload) Trip Feedback
- Isolator On Feedback
- Safety Interlock Feedback
- Motor Current Feedback (Analog Input) (Current above High Limit Setpoint)

PLC Interlocks

- Start Interlock Ready
- Process Interlock Ready
- Restart Delay Interlock
- There are no interlock faults latched

Motor Current – the motor current is monitored constantly after the running setpoint time has elapsed, if the current goes above the high current limit for five seconds the motor will trip on overcurrent.

Interlock fault recovery process – once a field interlock is activated the faceplate will display a red indication, if the interlock is recovered and still latched in the PLC function block and the faceplate will show a green indication and the reset button on the faceplate will flash, pressing on the reset button will unlatch the interlock and display a grey indication and allow the VSD to become ready for operation.

Maintenance mode – will disable the Start and Process Interlocks, but respects all the other interlocks. This is used to test a VSD while the process is stopped.

Simulation mode – will disable the physical output of the VSD and simulate an input run feedback after the elapsed setpoint time from the faceplate. The Start and Process Interlocks are respected. Simulation mode is used to test the process logic without starting the physical device itself. It is very useful in testing, commissioning and maintenance.

Start and stop requests are initiated by selecting a mode on the VSD faceplate:

- **Auto mode** - the auto control program initiates a start and stop, normal PLC control.
- **Manual mode & Desk mode** - the operator can start and stop from the faceplate.
- **Manual mode & Field mode** - the motor can be started and stopped from the start and stop push buttons in the field.

Typical Faceplate Graphics for the Advanced VSD Motor MAPS Template

VSD ISO	VSD MOTOR LEFT	VSD MOTOR RIGHT	VSD PUMP LEFT	VSD PUMP RIGHT
	VSD Reverse Direction Indication			
	VSD Forward Direction Indication			
	VSD Speed 01 Selected Indication			
	VSD Speed 02 Selected Indication			
	VSD Speed 03 Selected Indication			
	Manual Mode			
	Maintenance Mode			
	Simulation Mode			
	Motor Control Name			
Graphics legend:				
Line colour	Fill colour	VSD status		
Black	Grey	Stopped		
Grey	White	Stopping		
Green	White	Starting		
Black	Green	Running		
Black	Yellow/Red	Fault Active		

Signal Description		Agent Type	Advanced	Standard	Basic
SCADA Control Words:					
SSCL 0	Control Word	Marshal	x		
SSCL 1	Fail To Start Setpoint	Analog	x		
SSCL 2	Fail To Stop Setpoint	Analog	x		
SSCL 3	Running Setpoint	Analog	x		
SSCL 4	Stopping Setpoint	Analog	x		
SSCL 5	Simulate Setpoint	Analog	x		
SSCL 6	Restart Setpoint	Analog	x		
SSCL 7	Max Current Setpoint (High Limit)	Analog	x		
SSCL 8	Min In Current Setpoint (Raw Min)	Analog	x		
SSCL 9	Max In Current Setpoint (Raw Max)	Analog	x		
SSCL 10	Min Out Current Setpoint (Eng. Min)	Analog	x		
SSCL 11	Max Out Current Setpoint (Eng. Max)	Analog	x		

Signal Description		Agent Type	Advanced	Standard	Basic
SCADA Status Words:					
SSSL 0	Status Word	Marshal	x		
SSSL 1	Starting Time Process Value (actual)	Analog	x		
SSSL 2	Stopping Time Process Value (actual)	Analog	x		
SSSL 3	No of Operations (actual)	Analog	x		
SSSL 4	Restart Delay Time Remaining (actual)	Analog	x		
SSSL 5	Motor Current Process Value (actual)	Analog	x		
SSSL 6	Total Running Hours	Analog	x		
SSSL 7	Status Word 2	Marshal	x		

SCADA Control Word (Adroit Marshal Agent)		Event Logged	Advanced	Standard	Basic
Bit 0	Auto/Manual	x	x		
Bit 1	Field/Desk	x	x		
Bit 2	Maintenance	x	x		
Bit 3	Simulation	x	x		
Bit 4	Alarms All Reset		x		
Bit 5	Reset No Operations Accumulator	x	x		
Bit 6	Start Command	x	x		
Bit 7	Reset Running Hours Accumulator	x	x		
Bit 8	Stop Command	x	x		
Bit 9	Speed 01 Enabled	x	x		
Bit 10	Speed 02 Enabled	x	x		
Bit 11	Speed 03 Enabled	x	x		
Bit 12	Spare				
Bit 13	Spare				
Bit 14	Forward Command	x	x		
Bit 15	Reverse Command	x	x		

SCADA Status Word (Adroit Marshal Agent)		Alarmed	Advanced	Standard	Basic
Bit 0	Process Interlock		x		
Bit 1	Ready To Start		x		
Bit 2	Stopped		x		
Bit 3	Starting		x		
Bit 4	Running		x		
Bit 5	Stopping		x		
Bit 6	Fault		x		
Bit 7	Start Interlock OK		x		
Bit 8	Isolator Closed Fault	x	x		
Bit 9	Fault Fail To Stop	x	x		
Bit 10	Fault Fail To Start	x	x		
Bit 11	MCC Fault	x	x		
Bit 12	Safety Fault	x	x		
Bit 13	Overcurrent Fault	x	x		
Bit 14	Spare				
Bit 15	Forward Reverse Status Indication		x		

SCADA Status Word 2 (Adroit Marshal Agent)		Alarmed	Advanced	Standard	Basic
Bit 0	Speed 01 Active	x	x		
Bit 1	Speed 02 Active	x	x		
Bit 2	Speed 03 Active	x	x		
Bit 3	Spare				
Bit 4	Spare				
Bit 5	Spare				
Bit 6	Spare				
Bit 7	Spare				
Bit 8	Spare				
Bit 9	Spare				
Bit 10	Spare				
Bit 11	Spare				
Bit 12	Spare				
Bit 13	Spare				
Bit 14	Spare				
Bit 15	Spare				

Digital Inputs:		Advanced	Standard	Basic
DI 0	VSD Running Signal	x		
DI 1	Isolator Healthy (On = Healthy)	x		
DI 2	MCC Healthy (On = Healthy)	x		
DI 3	Safety Interlock Healthy (On = Healthy)	x		
DI 4	Manual Start Push Button Forward	x		
DI 5	Manual Start Push Button Reverse	x		
DI 6	Manual Stop Push Button (Off = Stop)	x		
Digital Outputs:				
DO 0	Device Start Forward	x		
DO 1	Device Start Reverse	x		
DO 2	Device Speed 01 Signal	x		
DO 3	Device Speed 02 Signal	x		
DO 4	Device Speed 03 Signal	x		
Analog Input:				
AI 0	Analog Input for Current Drawn	x		

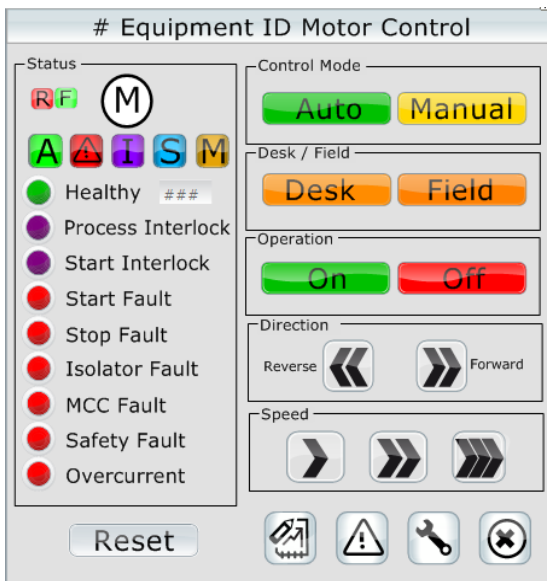
Resources	Advanced	Standard	Basic
Digital Inputs	7		
Digital Outputs	5		
Analog Inputs	1		
Analog Outputs	0		
PLC Memory Steps	643		
System Words Used	74		
System Bits Used	178		
SCADA SCAN Points	20		

Template: Advanced VSD PLC Function Block

<Device> (Advanced VSD Control)

FB_VSD_A_v1_0 VSD_A_v1_0			
DATA_VSD_A_V1_0.F_I_Run_FB	F_I_Run_FB	F_O_VSD_START_FWD_CMD	DATA_VSD_A_V1_0.F_O_VSD_START_FWD_CMD
DATA_VSD_A_V1_0.F_I_Isolator_Closed_FB	F_I_Isolator_Closed_FB	F_O_VSD_START_REV_CMD	DATA_VSD_A_V1_0.F_O_VSD_START_REV_CMD
DATA_VSD_A_V1_0.F_I_MCC_Healthy_FB	F_I_MCC_Healthy_FB	F_O_SPEED01	DATA_VSD_A_V1_0.F_O_VSD_SPEED01_CMD
DATA_VSD_A_V1_0.F_I_Safety_Interlock_OK	F_I_Safety_Interlock_OK	F_O_SPEED02	DATA_VSD_A_V1_0.F_O_VSD_SPEED02_CMD
DATA_VSD_A_V1_0.VAR_Process_Int_OK	VAR_Process_Int_OK	F_O_SPEED03	DATA_VSD_A_V1_0.F_O_VSD_SPEED03_CMD
DATA_VSD_A_V1_0.VAR_Start_Int_OK	VAR_Start_Int_OK	S_SW1_Marshall	DATA_VSD_A_V1_0.S_SW1_Marshall
DATA_VSD_A_V1_0.VAR_Auto_Run_FWD_Req	VAR_Auto_Run_FWD_Req	S_SW2_Starting_Time_PV	DATA_VSD_A_V1_0.S_SW2_Starting_Time_PV
DATA_VSD_A_V1_0.VAR_Auto_Run_REV_Req	VAR_Auto_Run_REV_Req	S_SW3_Stopping_Time_PV	DATA_VSD_A_V1_0.S_SW3_Stopping_Time_PV
DATA_VSD_A_V1_0.VAR_Auto_Speed01	VAR_Auto_Speed01	S_SW4_No_Of_Operations	DATA_VSD_A_V1_0.S_SW4_No_Of_Operations
DATA_VSD_A_V1_0.VAR_Auto_Speed02	VAR_Auto_Speed02	S_SW5_Restart_PV	DATA_VSD_A_V1_0.S_SW5_Restart_PV
DATA_VSD_A_V1_0.VAR_Auto_Speed03	VAR_Auto_Speed03	S_SW6_Current_PV	DATA_VSD_A_V1_0.S_SW6_Current_PV
DATA_VSD_A_V1_0.F_I_MAN_START_FWD_PB	F_I_MAN_START_FWD_PB	S_SW7_Running_Hours	DATA_VSD_A_V1_0.S_SW7_Running_Hours
DATA_VSD_A_V1_0.F_I_MAN_START_REV_PB	F_I_MAN_START_REV_PB	S_SW8_Marshall	DATA_VSD_A_V1_0.S_SW8_Marshall
DATA_VSD_A_V1_0.F_I_MAN_STOP_PB	F_I_MAN_STOP_PB	S_CW1_0_Auto_Manual	
DATA_VSD_A_V1_0.S_CW1_Marshall	S_CW1_Marshall	S_CW1_1_Desk_Field	
DATA_VSD_A_V1_0.S_CW2_Fail_To_Start_SP	S_CW2_Fail_To_Start_SP	S_CW1_2_Maintenance	
DATA_VSD_A_V1_0.S_CW3_Fail_To_Stop_SP	S_CW3_Fail_To_Stop_SP	S_CW1_3_Simulation	
DATA_VSD_A_V1_0.S_CW4_Running_SP	S_CW4_Running_SP	S_SW1_1_Ready_To_Start	
DATA_VSD_A_V1_0.S_CW5_Stopping_SP	S_CW5_Stopping_SP	S_SW1_2_Stopped	
DATA_VSD_A_V1_0.S_CW6_Simulate_SP	S_CW6_Simulate_SP	S_SW1_3_Starting_RQ	
DATA_VSD_A_V1_0.S_CW7_Restart_SP	S_CW7_Restart_SP	S_SW1_4_Running	
DATA_VSD_A_V1_0.S_CW8_MaxCurrent_SP	S_CW8_MaxCurrent_SP	S_SW1_5_Stopping_RQ	
DATA_VSD_A_V1_0.S_CW9_MinIn_Current_SP	S_CW9_MinIn_Current_SP	S_SW1_9_Fault_Stop	
DATA_VSD_A_V1_0.S_CW10_MaxIn_Current_SP	S_CW10_MaxIn_Current_SP	S_SW1_A_Fault_Start	
DATA_VSD_A_V1_0.S_CW11_MinOut_Current_SP	S_CW11_MinOut_Current_SP	S_SW1_D_Fault_OverCurrent	
DATA_VSD_A_V1_0.S_CW12_MaxOut_Current_SP	S_CW12_MaxOut_Current_SP		
DATA_VSD_A_V1_0.F_AI_Current	F_AI_Current_PV		

Template Graphic: Advanced Home Screen



Control	Descriptions
Auto	Auto control by PLC
Manual	Manual control by operator, desk and field mode
Desk	Manual control from the SCADA faceplate
Field	Manual control from the field start/stop station
On	Manual start the VSD
Off	Manual stop the VSD
Reset	Reset the VSD after a fault has been repaired
Healthy ###	Healthy to start, ### - Restart delay countdown in seconds
Process Interlock	VSD process ok to run interlock ready
Start Interlock	VSD ok to start interlock ready
Start Fault	VSD failed to start fault
Stop Fault	VSD failed to stop fault
Isolator Fault	VSD isolator off fault
MCC Fault	VSD MCC tripped fault
Safety Fault	VSD Safety off fault
Overcurrent	VSD overcurrent fault
Reverse	VSD reverse direction selection and indication
Forward	VSD forward direction selection and indication
Speed	VSD speed 1 to 3 selection and indication

Template Graphic: Advanced Setup Screen	Control	Descriptions
	Simulate	Enable simulation of VSD, disables the physical output
	Feedback Time	Time for simulation to simulate run feedback signal
	Maintenance	Enable maintenance mode (Disables all interlocks)
	Number of operations	Total number of VSD start operations
	Running Hours	Total number of VSD running hours
	Reset	Reset the total counter of the current service
	Start Signal Delay (SP)	Delay the PLC to accept the VSD is running
	Max Start (SP)	Time for running signal is on after start signal is on
	Last Start (PV)	Last time for running signal was on after start signal was on
	Stop Signal Delay (SP)	Delay the PLC to accept the VSD is stopped
	Max Stop (SP)	Time running signal is off after start signal is off
	Last Stop (PV)	Last time running signal was off after start signal was off
	Current Hi Limit	Sets the value of the maximum current for the VSD to trip at
	Current Value	Actual current the VSD is running at
	MinIn, MaxIn	Raw current input values for scaling
	MinOut, MaxOut	Engineering current values for scaling
	Restart (SP)	Restart prevention time of the VSD setpoint
	Restart (PV)	Remaining time before a VSD restart is allowed

Template Graphic: Advanced Alarms & Events Screen	Control	Descriptions
	Alarms	Display all the alarms filtered for the current Digital Output
	Events	Display all the events filtered for the current Digital Output.
	Local Acknowledge	Acknowledge all the alarms on locally on the current machine
	Global Acknowledge	Acknowledge all the alarms on globally on all the machines.

Template Graphic: Advanced Trend Screen	Control	Descriptions
	Current	Display the current of the VSD in the trend, with minimum, maximum and average values.
	Alarm Limit	Display the current Setpoint of alarming on high current of the VSD in the trend, with minimum, maximum and average values.